[0x05]

internal check

$f(\text{rax} \in [5,9]) \longrightarrow \{Y, N\}$
$\cup$
Bool

```
sub     rax, 5
cmp     rax, 4
```

sub rax, 5 ← sets min (normalizes) (of 5)

rax := rax − 5

rax == 5 → ZF = 1

CF = 0

~~cmp rax, 4~~

note: Sign flag (SF) := result negative

overflow flag (OF) := signed wrap-around

auxiliary flag (AF) := BCD (binary coded decimal)

cmp rax, 4 ← length of interval

ex) $4 \notin [5,9]$ SF = 1

rax = 4 − 5 = −1 & −1 < 4

$10 \notin [5,9]$

rax = 10 − 5 = 5 & 5 > 4

$6 \in [5,9]$

rax = 6 − 5 = 1 & 1 < 4

common jb ✓